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Question

Text 1

Chemically analyzing bulk rock-core samples from Australia's Mount McRae Shale, Ariel Anbar et al. found a transient but significant increase in molybdenum (Mo) at a point corresponding to roughly 2.5 billion years ago (Ga). On Earth, Mo is released mainly through oxidative weathering of minerals; Anbar et al. therefore concluded that atmospheric oxygen briefly increased 2.5 Ga, then returned to its earlier negligible level.

Text 2

Sarah Slotznick et al. reexamined the Mount McRae Shale core. Since chemically analyzing bulk samples can occlude contextual details, Slotznick et al. also employed high-resolution microscopy, which revealed volcanic debris—a known Mo host—around 2.5 Ga and microfractures in the surrounding matrix. The researchers assert that fluid could have reached the debris through the microfractures and initiated oxidative weathering long after debris deposition.

Based on the texts, Anbar et al. (Text 1) and Slotznick et al. (Text 2) would most likely disagree about the answer to which question about the portion of the Mount McRae Shale rock-core corresponding to 2.5 Ga?

Answer

- A. Does the increase in Mo in that portion suggest that atmospheric oxygen levels were not negligible before 2.5 Ga?
- B. Is the increase in Mo in that portion indicative of an increase in atmospheric oxygen dating to the same time?
- C. Is the increase in Mo in that portion attributable to oxidative weathering of the material found in that portion?
- D. Did chemically analyzing bulk samples lead to a false impression that there is an increase in Mo in that portion?